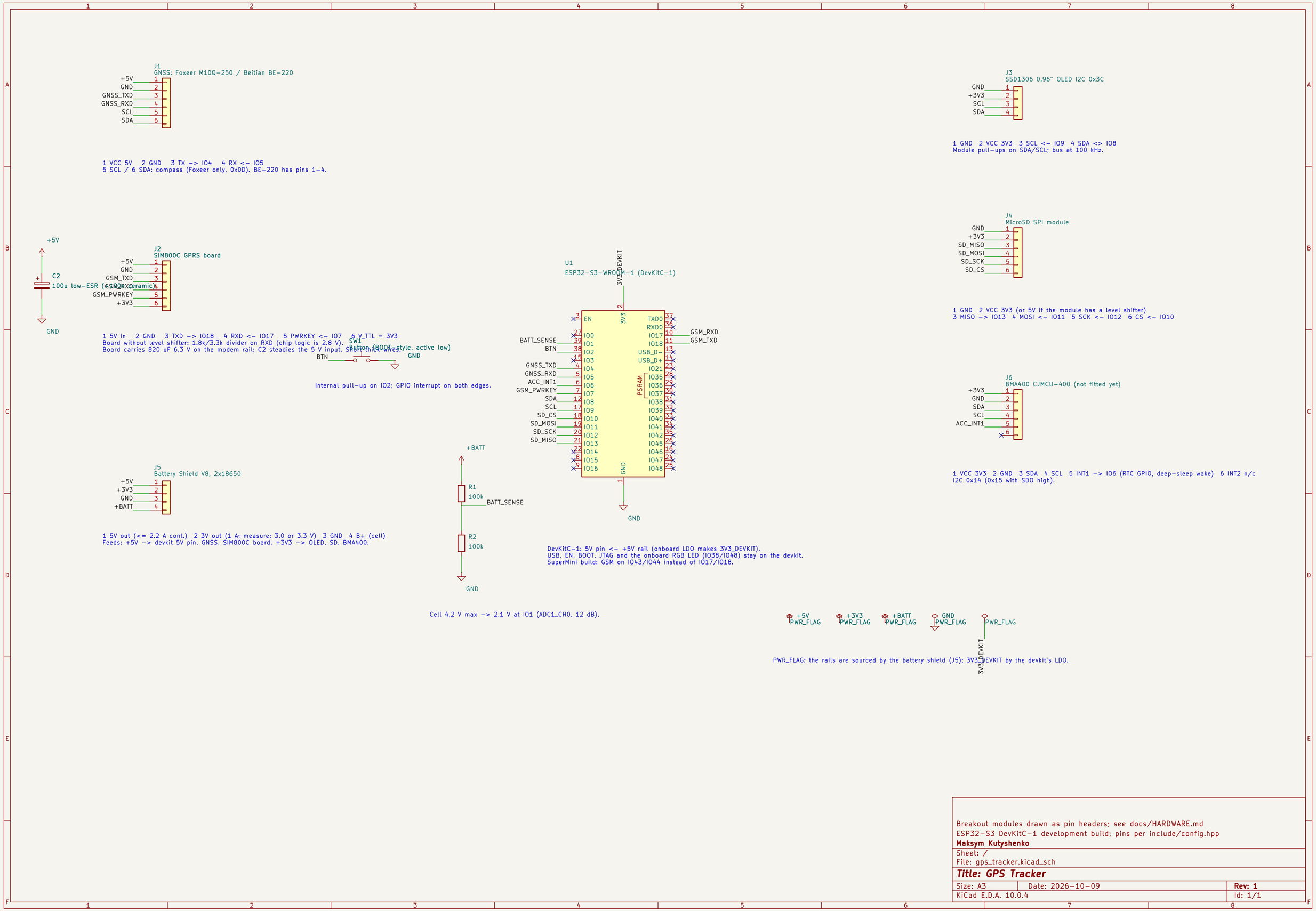

J1
GNSS: Foxeer M10Q-250 / Beitian BE-220
+5V
GND
GNSS_TXD
GNSS_RXD
SCL
SDA
1 VCC 5V 2 GND 3 TX -> IO4 4 RX <- IO5
5 SCL / 6 SDA: compass (Foxeer only, 0x0D). BE-220 has pins 1-4.
J3
SSD1306 0.96" OLED I2C 0x3C
GND
+3V3
SCL
SDA
1 GND 2 VCC 3V3 3 SCL <- IO9 4 SDA <> IO8
Module pull-ups on SDA/SCL; bus at 100 kHz.
J4
MicroSD SPI module
GND
+3V3
SD_MISO
SD_MOSI
SD_SCK
SD_CS
1 GND 2 VCC 3V3 (or 5V if the module has a level shifter)
3 MISO -> IO13 4 MOSI <- IO11 5 SCK <- IO12 6 CS <- IO10
+5V
C2
100u low-ESR (≥ 6.3 V ceramic)
GND
J2
SIM800C GPRS board
+5V
GND
GSM_TXD
GSM_RXD
GSM_PWRKEY
+3V3
1 5V in 2 GND 3 TXD -> IO18 4 RXD <- IO17 5 PWRKEY <- IO7 6 V_TTL = 3V3
Board without level shifter: 1.8k/3.3k divider on RXD (chip logic is 2.8 V).
Board carries 820 uF 6.3 V on the modem rail; C2 steadies the 5 V input. Short thick wires.
SW1
Button (BOOT-style, active low)
BTN
GND
Internal pull-up on IO2; GPIO interrupt on both edges.
U1
ESP32-S3-WROOM-1 (DevKitC-1)
3V3_DEVKIT
EN
3V3
GND
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
BATT_SENSE
BTN
GNSS_TXD
GNSS_RXD
ACC_INT1
GSM_PWRKEY
SDA
SCL
SD_CS
SD_MOSI
SD_SCK
SD_MISO
GSM_RXD
GSM_TXD
J6
BMA400 CJMCU-400 (not fitted yet)
+3V3
GND
SDA
SCL
ACC_INT1
1 VCC 3V3 2 GND 3 SDA 4 SCL 5 INT1 -> IO6 (RTC GPIO, deep-sleep wake) 6 INT2 n/c
I2C 0x14 (0x15 with SDO high).
J5
Battery Shield V8, 2x18650
+5V
+3V3
GND
+BATT
1 5V out (<= 2.2 A cont.) 2 3V out (1 A; measure: 3.0 or 3.3 V) 3 GND 4 B+ (cell)
Feeds: +5V -> devkit 5V pin, GNSS, SIM800C board. +3V3 -> OLED, SD, BMA400.
+BATT
R1
100k
BATT_SENSE
R2
100k
GND
Cell 4.2 V max -> 2.1 V at IO1 (ADC1_CH0, 12 dB).
DevKitC-1: 5V pin <- +5V rail (onboard LDO makes 3V3_DEVKIT).
USB, EN, BOOT, JTAG and the onboard RGB LED (IO38/IO48) stay on the devkit.
SuperMini build: GSM on IO43/IO44 instead of IO17/IO18.
+5V
PWR_FLAG
+3V3
PWR_FLAG
+BATT
PWR_FLAG
GND
PWR_FLAG
PWR_FLAG
3V3_DEVKIT
PWR_FLAG: the rails are sourced by the battery shield (J5); 3V3_DEVKIT by the devkit's LDO.
Breakout modules drawn as pin headers; see docs/HARDWARE.md
ESP32-S3 DevKitC-1 development build; pins per include/config.hpp
Maksym Kutyshenko
Sheet: /
File: gps_tracker.kicad_sch
Title: GPS Tracker
Size: A3
Date: 2026-10-09
Rev: 1
KiCad E.D.A. 10.0.4
Id: 1/1